Phenological sensitivity to climate change disrupts species' coexistence

D.M. Buonaiuto [1,2,3,a], M. Donahue[4], E.M. Wolkovich[2,3,5]

June 30, 2026

Author affiliations:

[1]Department of Plant Science and Landscape Architecture , University of Maryland, College Park, Maryland, USA. ORCID: 0000-0003-4022-2591

[2]Arnold Arboretum of Harvard University, Boston, Massachusetts, USA.

[3]Department of Organismic and Evolutionary Biology, Harvard University, Cambridge, Massachusetts, USA

[4]Hawai'i Institute of Marine Biology, University of Hawai'i at Manoa, Kan'eohe, HI, USA.

[5]Forest & Conservation Sciences, Faculty of Forestry, University of British Columbia, Vancouver, British Columbia, Canada

[a]Corresponding author: 617.823.0687; dbuona@umd.edu

1 Abstract

Shifts in phenology with climate change are likely to restructure communities by changing species interactions, but current frameworks cannot forecast these impacts. Here we introduce a flexible and simple approach—explicitly considering phenology as the outcome of an interaction between the environment and species' unique physiology—that could rapidly advance community forecasting. Using this approach, we show how phenological-mediated patterns of coexistence under a historical climate regime may degrade to competitive exclusion with climate change.

2 Main

Each spring, in ecosystems across the temperate zone, ecological communities re-assemble from their period of dormancy with the hatching of insects, arrival of migratory birds, and emergence of new seedlings, leaves and flowers. The seasonal timing of these life cycle events—their phenology—can be quite different, even among closely related organisms in shared habitats (1; 2). These phenological differences have often been overlooked in ecological theory outside of plant-pollinator interactions, but growing evidence suggests they may play a major role in shaping interactions and ecological communities (3; 4; 5).

Experiments show that the order in which species begin their seasonal growth can alter the outcomes of competitive interactions (6; 7). Such phenological differences among species can generate seasonal priority effects where earlier active species gain a competitive advantage (8; 9). Studies in plants (10; 11), amphibians (12), insects (13; 14) and birds (15) have found that apparently small seasonal priority effects, on the order of just a few days, can alter competitive hierarchies in ways that persist across several growing seasons (16).

The importance of seasonal priority effects in mediating interspecific competition suggests how phenology may be fundamental to coexistence (17; 4), but not how phenological differences between species arise. Functionally, phenology is the realized product of an interaction between the environment (cues) and a species' unique physiology, genetic and developmental attributes (18). Differences in how species respond to the same environmental cues are one of the primary mediators of the trade-off between phenology and competition. With environmental cues changing rapidly in response to anthropogenic warming, phenological shifts have become one of the most pronounced indicators of climate change to date. (19). Understanding the degree to which a species responds to environmental variation, known as phenological sensitivity (Δ phenology/ Δ environmental cue), is key to predicting community assembly in an era of global change.

Environmental cues such as temperature, light and moisture control phenology for most species (20), with many responding to multiple aspects of cues (21; 22). For example, temperate plants that rely on temperature as a cue often require exposure to both cool winter (0-4°C, hereafter: winter chilling, which is similar to cold stratification) followed by warm spring temperatures to initiate phenological transitions (23). Controlled experiments have helped quantify species-specific responses to cues, which—when combined with the exact environmental conditions—drive differences in realized phenology between species.

Realized phenology can vary widely among species $\times$ environment (e.g., 24). We illustrate with using germination data from experiments for 11 co-occurring species (25) that show very different sensitives to the same environmental cues (Figure 1a). These differences could impact competition, especially as they can determine which species germinates first depends on environmental conditions, and the order switches with increasing cue levels (for one example pair, see Figure 1b,c). This illustrates that the strength of seasonal priority effects are highly dependent on the environment, yet this is not how most coexistence models approach phenology.

Recent assembly models have begun to consider phenology (5; 8; 9), including the potential for seasonal priority effects, but do not include realized phenology. Instead these models center on the relationship between timing and competition, generally including a trade-off between the two (5; 8; 9). This trade-off has been expressed explicitly through a connection of phenological differences to parameters controlling the strength of competition, or implicitly by providing a fitness advantage to earlier species offset by poorer performance under low resource conditions (e.g., R^* in 8; 9).

Models providing a fitness advantage to earlier species all recognize the importance of environmental variability, generally including interannual variation in the environment that in turn drives differences in phenology. Many models—focused often on grasslands and water competition—have assigned phenology as equivalent to an intrinsic species-level timing (or season length strategy). Year-to-year variation in the environment then alters absolute differences between species, which consistently co-varies with resource competition (26; 9). In these approaches the relative orders between species remain fixed, but other models allow species relative orders and timings to vary year-to-year assuming the underlying relationship between pairwise differences in timings and interspecific-competition is known—and static (5). Each of these approaches thus fixes one aspect of what is actually a dynamic environment $\times$ phenology $\times$ competition relationship. This is an understandable simplification, but one that may make forecasting community shifts with climate change challenging, as the environment becomes more variable.

Phenological models of community assembly have included environmental variability (5; 9) that assumes an underlying stationary environment (one that varies year-to-year but with an underlying constant distribution), in contrast to the reality of climate change. Because environmental variability in assembly models structures what species combinations are stable, shifts in the environment with anthropogenic climate change are likely to reshape communities, but forecasting them requires accurately capturing how the environment affects phenology and competition together in a simple—but still robust—form. This means capturing how the environment can determine both the absolute (e.g., earlier in the year) and relative (e.g., order of arrival) timings of species, and the how these timings impact coexistence. While current models to date have simplified one aspect of this problem, it may be possible to address it through new approaches.

We adapt a common coexistence model to address these limitations: including inter- and intra-annual environmental variability then modeling phenology more physiologically to better capture how competition depends on the environment filtered through species phenological sensitivities. Our approach includes phenological sensitivity in its simplest form—two species with differential sensitivity to one environmental cue. This approach is similar to others in varying species intrinsic competitive abilities at the same time we varied phenology, but differed in modeling phenological sensitivities alongside intra-annual environmental variation. Any changes in the environment thus filter through species-specific sensitivities, and transform differences in realized phenology between species (Figure 2b).

Comparing long-term co-occurrence (which we consider coexistence here) of species pairs under two different climate regimes, we found that no pair coexisting under one climate could coexist under the other (Figure 2a). Thus, for any species that currently coexist due to the tradeoff between relative phenology and intrinsic competitive ability, climate change disrupts this coexistence mechanism.

These results highlight that improving the predictive power of coexistence models with climate change likely requires rethinking how we measure species phenology. We found that the required tradeoff between differences in realized phenology and competitive differences is stable across a shifting environment (i.e., the same magnitude of phenological differences are required to offset the same level of differences in competitive ability), and thus uninformative on its own for predicting which species pairs can coexist. Instead, it is the tradeoff between phenological sensitivity and competition differences that change in a novel environment (Figure 2b,c). Thus, measuring species phenologies may be far less useful for understanding future community assembly than measuring their phenological sensitivities.

Phenological sensitivity is a trait that can be readily measured for real species, providing a link between assembly theory and observed ecological changes. More studies that quantify species-level phenological sensitivity could thus improve forecasting. Studies that estimate the phenological sensitivity to multiple environmental cues could address how much local adaptation versus plasticity structure phenological sensitivity and competitive traits (e.g. (27; 28), but will be most useful for forecasting if they include a greater scope of taxa and lifeforms. While much ecological work has focused on annual plants, herbaceous perennials and woody plants are the dominant functional types in many ecological systems.

Improved forecasting will also require assembly models that better incorporate the cues that underlie phenology. Exactly how best to include these cues should likely be determined by one of the main factors that selects for them—environmental risk. For example, leafout of many forest trees relies on cues of winter chilling and/or photoperiod, which could reduce the risk of frost damage in unstable spring environments. Introducing biologically realistic mortality events into community assembly models that represent important selective forces, such as late-season frosts or early droughts, offers a straightforward approach to assess the importance of this. Ecological models generally subsume such risks into species traits (e.g., increasing maintenance costs), which misses the reality that such risks are also product of timing $\times$ environment. Such approaches would translate ecological theory into practice and advance a more predictive ecology than is better equipped to inform ecosystem conservation and management.

Figures

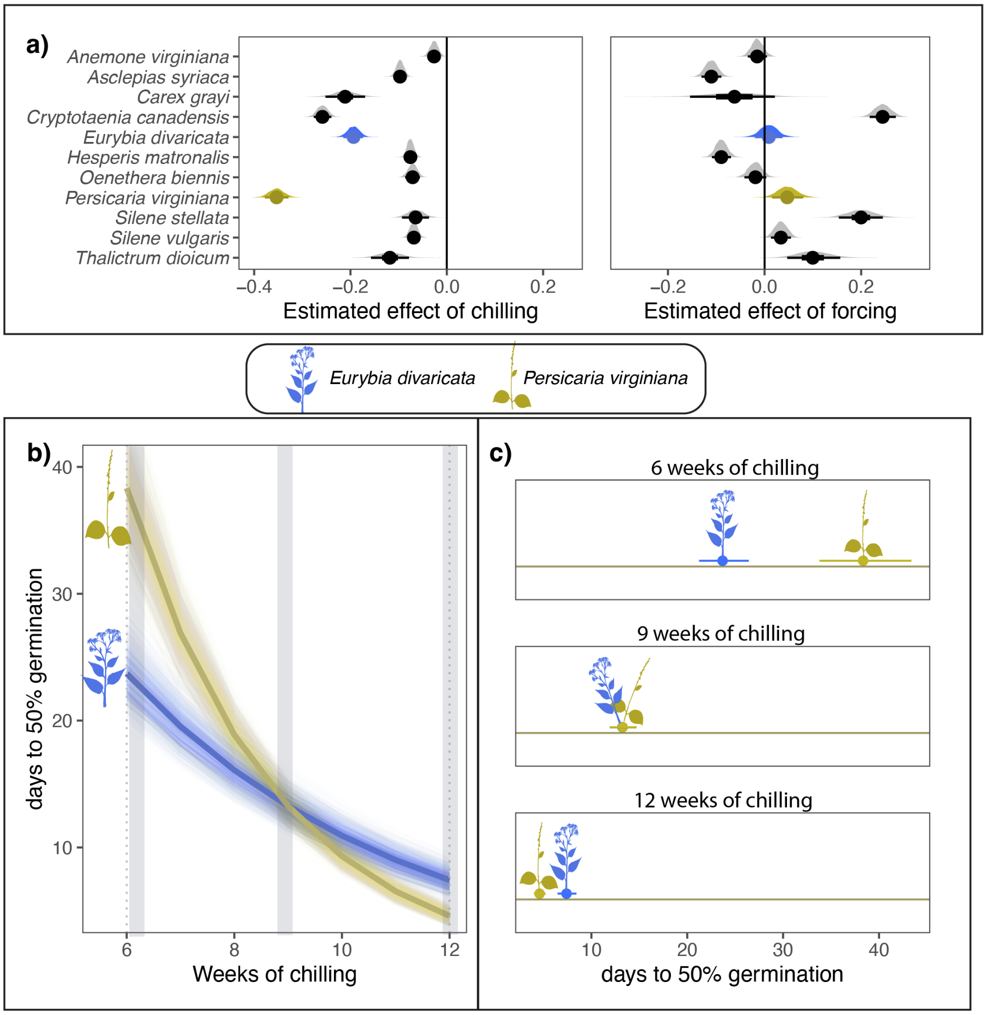

Figure 1: **Co-occurring species can differ substantially in their phenological sensitivity (Δ phenological event/Δ climate) to the environment.** Panel a) estimates the phenological sensitivity of germination timing to winter chilling and spring forcing for eleven herbaceous species of eastern North America. Points represent the mean estimate and thick and thin bars represent the 50 and 95% uncertainty intervals (with posteriors shown above lines as gray distributions; for details on the model, see the Supporting Information). In b) and c) we highlight the phenological sensitivity to winter chilling of two of these species—*Eurybia divaricata* and *Persicaria virginiana*—that are commonly observed in competition. b) Depicts the sensitivity of each species to winter chilling, with thick lines representing the mean estimate and lighter lines representing modeled uncertainty. Panel c) illustrations how the species' different sensitivities manifest in contrasting realized germination differences under different climate conditions.

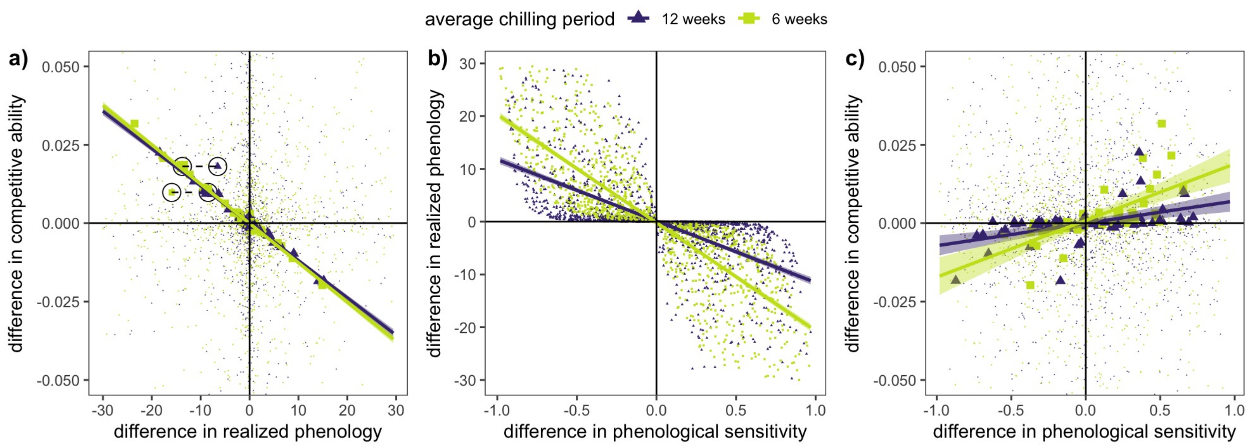

Figure 2: **Climate change will alter the relationships between competitive ability, phenological sensitivity, realized phenology and coexistence.** Panel a) depicts the trade-off between differences in realized phenology and differences in competitive abilities (in all panels each point represents a species pair with randomly selected phenological sensitivities and competitive abilities subjected to two climate scenarios—12 and 6 weeks of chilling; species pairs that co-occurred after 300 years are in bold). The trade-off needed to maintain coexistence does not change between climate scenarios, but no species pairs that coexisted under one scenario can coexist under the other scenario, as highlighted by the two sets of species pairs circled in panel a). Panel b) shows the relationship between differences in phenological sensitivity and differences in realized phenology, which depend on the environment. Panel c) depicts the trade-off between differences in phenological sensitivity and differences in competitive ability under the two climate scenarios (see Supporting Information for details).

References

[1] Ovaskainen, O., Skorokhodova, S., Yakovleva, M., Sukhov, A., Kutenkov, A., Kutenkova, N., Shcherbakov, A., Meyke, E., and del Mar Delgado, M. *Proceedings of the National Academy of Sciences* **110**(33), 13434–13439 (2013).

[2] An, S., Chen, X., Shen, M., Zhang, X., Lang, W., and Liu, G. *Frontiers in Plant Science* **Volume 13 - 2022** (2022).

[3] Cleland, E. E. and Wolkovich, E. *Annual Review of Ecology, Evolution, and Systematics* **55**(Volume 55, 2024), 471–492 (2024).

[4] Blackford, C., Germain, R. M., and Gilbert, B. *The American Naturalist* **195**(6), E168–E180 (2020). PMID: 32469665.

[5] Rudolf, V. H. W. *Ecology Letters* (2019).

[6] Dawson-Glass, E., Schiafo, R., Kuebbing, S. E., and Stuble, K. L. *Ecology* **106**(1), e4504 2025/07/15 (2025).

[7] Buonaiuto, D. M. and Wolkovich, E. M. *Journal of Ecology* **111**, 1711–1721 2023/07/19 (2023).

[8] Wolkovich, E. and Donahue, M. J. *Biological Reviews* **96**(6), 2810–2827 (2021).

[9] Levine, J. I., Levine, J. M., Gibbs, T., and Pacala, S. W. *Ecology Letters* **25**(5), 1110–1125 (2022).

[10] Wainwright, C. E., Wolkovich, E. M., and Cleland, E. E. *Journal of Applied Ecology* **49**(1), 234–241 Nov (2011).

[11] Koerner, C., Stoecklin, J., Reuther-Thiebaud, L., and Pelaez-Riedl, S. *New Phytologist* **177**(3), 698–705 (2008).

[12] Rudolf, V. H. W. and Singh, M. *Oecologia* **173**(3), 1043–1052 (2013).

[13] Ekholm, A., Tack, A. J. M., Pulkkinen, P., and Roslin, T. *Journal of Animal Ecology* **89**(3), 829–841 (2020).

[14] Rasmussen, N. L., Van Allen, B. G., and Rudolf, V. H. W. *Journal of Animal Ecology* **83**(5), 1206–1215 (2014).

[15] Samplonius, J. M. and Both, C. *Current Biology* **29**(2), 327–331.e2 2025/07/15 (2019).

[16] Vaughn, K. J. and Young, T. P. *Ecological Applications* **25**(3), 791–799 2022/03/02/ (2015).

[17] Diez, J., Schlauch, J., DuCharme, E., Collings, J., and Erskine, S. *Journal of Ecology* **n/a**(n/a).

[18] Donohue, K., DORN, L., Griffith, C., Kim, E., Anna, A., Polisetty, C. R., and Johanna, S. *Evolution* **59**(4), 758–770 04 (2005).

[19] Canadell, J., Monteiro, P., Costa, M., Cotrim da Cunha, L., Cox, P., Eliseev, A., Henson, S., Ishii, M., Jaccard, S., Koven, C., Lohila, A., Patra, P., Piao, S., Rogelj, J., Syampungani, S., Zaehle, S., and Zickfeld, K. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, New York, NY, (2021).

[20] Forrest, J. and Miller-Rushing, A. J. *Philosophical Transactions of the Royal Society B: Biological Sciences* **365**(1555), 3101–3112 Sep (2010).

[21] Lang, W., Zhang, Y., Li, X., Meng, F., Liu, Q., Wang, K., Xu, H., Chen, A., Peñuelas, J., Janssens, I. A., and Piao, S. *Nat Ecol Evol* **9**(2), 261–272 Feb (2025).

[22] Chuine, I. and Régnière, J. *Annual Review of Ecology, Evolution, and Systematics* **48**(Volume 48, 2017), 159–182 (2017).

[23] Ettinger, A. K., Chamberlain, C. J., Morales-Castilla, I., Buonaiuto, D. M., Flynn, D. F. B., Savas, T., Samaha, J. A., and Wolkovich, E. M. *Nature Climate Change* **10**(12), 1137–1142 (2020).

[24] Fernández-Pascual, E., Carta, A., Mondoni, A., Cavieres, L. A., Rosbakh, S., Venn, S., Satyanti, A., Guja, L., Briceño, V. F., Vandelook, F., Mattana, E., Saatkamp, A., Bu, H., Sommerville, K., Poschlod, P., Liu, K., Nicotra, A., and Jiménez-Alfaro, B. *New Phytologist* **229**(6), 3573–3586 (2021).

[25] Buonaiuto, D. M. and Wolkovich, E. M. (2023).

[26] Godoy, O. and Levine, J. M. *Ecology* **95**(3), 726–736 (2014).

[27] Lipowsky, A., Roscher, C., Schumacher, J., Michalski, S. G., Gubsch, M., Buchmann, N., Schulze, E.-D., and Schmid, B. *Perspectives in Plant Ecology, Evolution and Systematics* **17**(1), 66–77 (2015).

[28] Liang, L. and Wu, J. *International Journal of Biometeorology* **65**(11), 1953–1966 (2021).